

VR Mosque – Immersive Virtual Reality Environment

Project Overview

VR Mosque is an immersive virtual reality environment developed using Unity, C#, and Blender. The project focuses on creating a detailed and interactive mosque interior, combining custom 3D modeling, architectural design, environmental lighting, visual effects, and VR interactions.

The mosque's exterior structure was purchased as a third-party asset, while the interior environments and their 3D assets were designed and developed from scratch.

My primary contributions included creating custom architectural and decorative assets, developing stained-glass windows and their visual effects, designing interior spaces, and implementing an interactive water simulation for the ablution (Wudu) area.

Development and Implementation

1. Mosque Environment and Interior Design

- Integrated the purchased mosque exterior model into the Unity environment.
- Designed and constructed the interior layouts to complement the existing architectural structure.
- Created custom 3D assets for the interior, including furniture, decorative objects, architectural details, and environmental props.
- Developed detailed interior spaces incorporating wooden flooring, structural beams, walls, windows, and ceiling decorations.
- Applied materials and textures to create visually consistent architectural environments.
- Organized and positioned assets within Unity to maintain appropriate proportions and spatial relationships.

2. Custom 3D Asset Modeling

Created the interior 3D assets used throughout the mosque environment.

Implementation:

- Modeled furniture, architectural components, and decorative objects for the interior spaces.

- Developed custom environmental props, including tables, seating arrangements, and food-related objects.
- Designed assets to fit the scale and visual style of the mosque.
- Integrated the completed models into Unity and configured their materials and placement.
- Assembled individual assets into fully furnished interior environments.

This process involved both individual asset creation and complete 3D scene composition.

3. Stained-Glass Window Design and Lighting Effects

Designed and developed custom stained-glass windows to enhance the mosque's architectural appearance and interior atmosphere.

Implementation:

- Created custom stained-glass window designs and integrated them into the mosque's interior architecture.
- Developed the visual appearance of the glass using Unity materials and lighting effects.
- Configured the stained-glass effect to simulate the appearance of light interacting with decorative glass surfaces.
- Integrated the windows with the surrounding architectural elements.
- Adjusted the visual presentation to complement the interior lighting and environment.

Technical Focus:

The stained-glass implementation combined custom asset design, material configuration, and environmental lighting to create a decorative visual effect within the mosque.

4. Ablution (Wudu) Area Development

Designed and developed a dedicated ablution environment containing washing stations and a custom interactive water system.

Environment Construction:

- Created the architectural layout of the washing area.

- Developed the surrounding interior elements, including tiled surfaces, decorative wall patterns, and washing stations.
- Modeled and arranged environmental components to create a consistent and functional-looking space.
- Integrated the washing stations into the mosque's overall interior environment.

5. Interactive Water Simulation Using Unity Particle Systems

Developed a custom water effect using **two overlapping particle emission systems** within Unity.

Implementation:

- Created two separate particle emission systems to represent flowing water.
- Positioned the emission systems to overlap and combine their visual output.
- Used the layered particle emissions to produce a more continuous and visually detailed water stream.
- Adjusted the placement of the particle systems to align with the washing station.
- Integrated the water effect into the ablution environment as an interactive feature.

Technical Approach:

Instead of relying on a single particle emitter, the water effect was developed by combining two overlapping particle systems. The layered emissions work together to simulate flowing water, adding visual depth to the effect.

This implementation demonstrates practical experience with Unity's Particle System, environmental visual effects, and interactive VR development.

6. Scene Assembly, Materials, and Lighting

- Integrated custom 3D assets into the Unity environment.
- Configured materials and textures for architectural and decorative objects.
- Developed and adjusted environmental lighting to improve the appearance of interior spaces.
- Integrated the stained-glass visual effects with the surrounding environment.
- Organized the scene hierarchy to manage architectural components, furniture, lighting, and particle systems.

- Combined the interior environments and purchased exterior into a unified VR scene.

Technologies Used

Technology	Application
Unity	VR environment development, scene assembly, and lighting
C#	Application scripting and interaction development
Blender	Custom 3D asset modeling
Unity Particle System	Interactive water simulation
Unity Materials	Stained-glass and surface visual effects
XR Interaction Toolkit	VR interaction development

My Contributions

- Designed and created the custom interior 3D assets.
- Developed the mosque's interior architectural environments.
- Created furniture, decorative elements, and environmental props.
- Designed and implemented stained-glass windows and their visual effects.
- Developed the ablution area and its washing stations.
- Implemented an interactive water simulation using two overlapping particle emission systems.
- Integrated assets, materials, lighting, and visual effects into Unity.
- Collaborated on the overall development and integration of the VR environment.

Third-Party Assets

The mosque's exterior model was purchased as a third-party asset. The interior environments, custom 3D assets, stained-glass effects, and water simulation were developed as part of the project.







